LAB 08

Execution Stage: an Order Manager that Trades the Paper Account in Any Session

Item	Detail
Topic	Topic 2: Technical Indicators, Trading Strategies and Metrics with TradingView and Alpaca
Pipeline stage	Execution
Duration	25 min
Simulator page	http://localhost:8000/#/trading?tab=ticket
Mode	No coding. Runs ONLINE with your Alpaca paper keys (Integrations page), or OFFLINE on the demo data and the built-in paper broker. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

The pipeline now produces target positions. The desk needs an order manager that turns 'target' versus 'actual' into paper orders safely: the right order type for the current US session (the Singapore class runs while New York is in its overnight and pre-market sessions), prices off the live quote, no duplicate orders if a job runs twice, and a printed order lifecycle the trader can match in the Alpaca dashboard.

Objective

Place market, limit, stop and bracket paper orders, including an extended-hours limit order, and follow each order's lifecycle.

Before You Start

- Make sure the Trading Simulator is running and open http://localhost:8000 > Course Labs > Lab 08. It is installed once - see the guide 'Download, Install and Start the Trading Simulator Software' in the Learner Guide.
- Record every result you are asked to check - you will use it in the debrief and in the later labs.

Lab Steps - In the Trading Simulator

No coding is needed. Make sure the Trading Simulator is running (installed once - see the guide 'Download, Install and Start the Trading Simulator Software') and open http://localhost:8000. In the sidebar open Course Labs > Lab 08: the app shows the same steps with a Go button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.

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Step 1. Market order in the regular session

Open Paper Trading > Order ticket, fill in the fields below and press Send paper order. A market order buys straight away at the current price; it is accepted in the regular session.

- **Account:** Default paper account - the simulator's built-in paper broker (not TradingView). Keep it for every step: Session (sim) works only on this account, while the Alpaca paper account always uses the real clock.
- **Symbol:** SPY
- **Side:** buy
- **Qty:** 10
- **Type:** market
- **Time in force:** day
- **Limit price, Stop price:** leave blank
- **extended hours:** leave unticked
- **Bracket take-profit, Bracket stop-loss:** leave blank
- **Session (sim):** regular - pretends New York is in its regular session (21:30-04:00 Singapore time, 22:30-05:00 in winter); 'real clock' uses the actual time

Open: http://localhost:8000/#/trading?tab=ticket (Paper Trading > Order ticket)

Check: The Result panel shows ACCEPTED - all checks passed, 'BUY 10 SPY MARKET DAY -> filled', and each risk check (kill_switch, duplicate, session, allow_list, long_only, daily_loss, max_order_qty, max_order_value, max_position_pct) marked ok.

Step 2. Extended-hours limit order (Singapore class hours)

Keep the Step 1 ticket and change only the fields below, then press Send paper order. Session (sim) = overnight pretends New York is in its overnight session - where it is during Singapore class hours. Only LIMIT orders are accepted then, so 'auto' builds one for you: time in force DAY, extended hours on, and a limit price 0.25% (25 bps) above the ask for a buy, which caps the price you pay. Then send a market order in the same session to see it blocked.

- **Type:** auto (session-aware)
- **Session (sim):** overnight
- **Limit price:** leave blank - auto sets it from the quote
- **extended hours:** leave unticked - auto switches it on
- **Everything else:** as in Step 1
- **Second send:** Type = market, Session (sim) still overnight

Open: http://localhost:8000/#/trading?tab=ticket

Check: First send: ACCEPTED, 'BUY 10 SPY LIMIT DAY extended_hours limit <price> -> filled'; the Orders tab shows 'limit day ext-hrs', session overnight. Second send (market): BLOCKED - the session check reads 'overnight session accepts only LIMIT orders with extended_hours=True'.

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Step 3. Bracket order

A bracket order is an entry plus two automatic exits: a take-profit (a sell limit above the price, to lock in a gain) and a stop-loss (a sell stop below it, to cap a loss). Brackets are accepted in the regular session only, and both exit prices must be filled in. Fill in the fields below and press Send paper order, then open the Orders tab and click each status badge to see that order's lifecycle.

- **Qty:** 5
- **Type:** market
- **Session (sim):** regular
- **Bracket take-profit:** about 3% above the current SPY price - e.g. SPY at 770 -> 793.10 (read the price in the Fill column of your Step 1 order)
- **Bracket stop-loss:** about 2% below it - e.g. SPY at 770 -> 754.60
- **Limit price, Stop price:** leave blank
- **extended hours:** leave unticked
- **Everything else:** as in Step 1

Open: http://localhost:8000/#/trading?tab=ticket

Check: Orders tab: the parent 'buy 5 SPY market day bracket' is filled, and two sell legs appear with status new - 'limit gtc bracket (leg)' at the take-profit and 'stop gtc bracket (leg)' at the stop-loss. They were held until the parent filled; when one leg fills, the other is cancelled (one-cancels-other).

Step 4. Queue while closed and cancel

Session (sim) = closed pretends the US market is shut (weekends and US holidays). 'auto' turns the order into a GTC (good-till-cancelled) limit order that queues for the next session instead of filling. First clear the two bracket fields - the ticket keeps them from Step 3. Press Send paper order, then open the Orders tab and cancel the queued order.

- **Qty:** 10
- **Type:** auto (session-aware)
- **Session (sim):** closed
- **Bracket take-profit, Bracket stop-loss:** clear both (blank)
- **Limit price:** leave blank - auto sets it
- **Time in force:** leave as day - auto changes it to gtc
- **Everything else:** as in Step 1
- **Then:** Orders tab > the new 'limit gtc' row (status accepted) > press the red cancel at the right-hand end

Open: http://localhost:8000/#/trading?tab=ticket

Check: Result: ACCEPTED, 'BUY 10 SPY LIMIT GTC limit <price> -> accepted', and the session check reads 'market closed - GTC limit order will queue for the next session'. After cancel the status is canceled; click the badge to see new -> accepted -> canceled. Tip: while the real US market is open (21:30-04:00 Singapore time) the simulator's background market can fill a queued order within a minute - cancel straight away, or send it again if it already filled.

Expected in the app: A blotter with filled, working, queued and cancelled orders and the risk checks for each.

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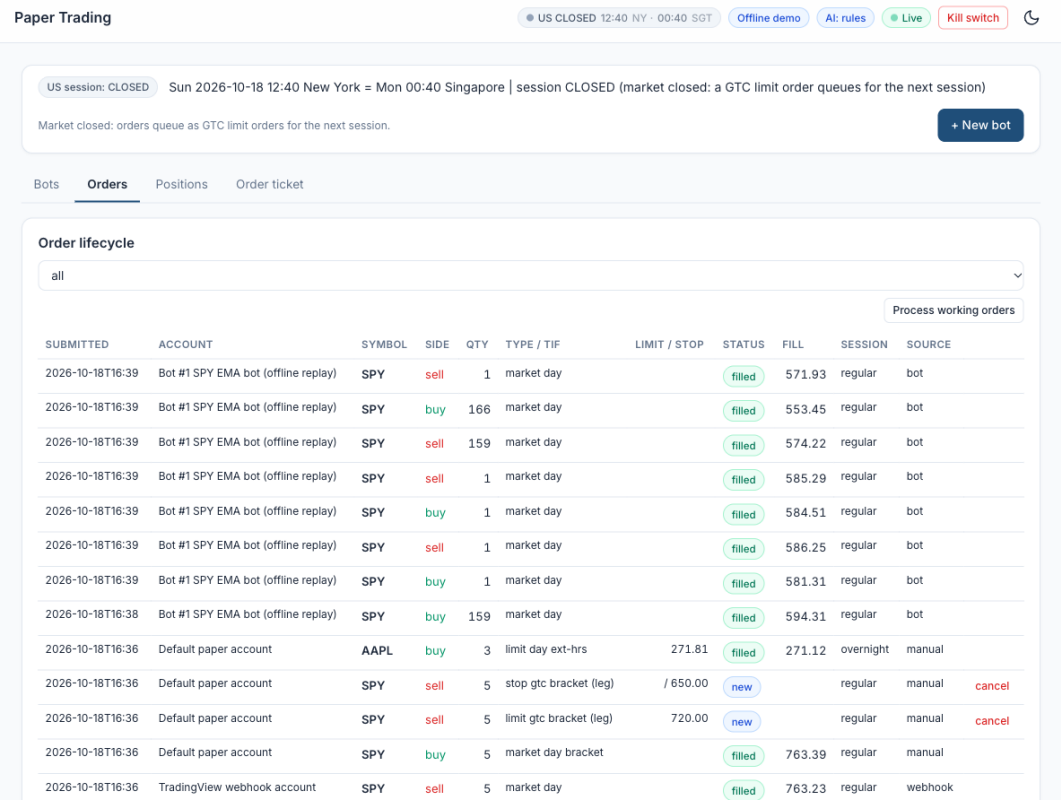


Figure: Lab 08 in the Trading Simulator.

Optional: Let AI Write the Code

This lab needs no coding. To see, run or customise the code behind it, click a link: the AI opens with the prompt filled in and writes the code for you. All prompts are also in labs/Prompt_Library.html and .pdf.

Place paper orders on Alpaca - run it in Google Colab
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: using alpaca-py with a PAPER account (TradingClient(..., paper=True)), read my keys from Google Colab Secrets named APCA_API_KEY_ID and APCA_API_SECRET_KEY (never put keys in the code), then place: a market order to buy 1 SPY; a limit order to buy 1 SPY 2% below the last price with extended_hours=True and time in force DAY; and a bracket order with a 3% take-profit and a 2% stop-loss. Print each order's id and status, list the open orders, then cancel the limit order.

Open in Claude | Open in ChatGPT | Open Google Colab

Guide: Paper Trading on Alpaca, Step by Step - Manual and Automated

First place one order by hand to see the order lifecycle, then let the simulator place orders automatically (Labs 8 and 16).

1. Place a manual paper order

In the Alpaca dashboard select the Paper Trading account (upper left), search the symbol (for example SPY) and open its page. Set the side, quantity, order type and time in force, then press Buy.

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2. Follow the order lifecycle

Open the Orders view and watch the status move from new / accepted to filled (or cancelled). Open Positions to see the holding, cost basis and unrealised P&L.

3. Know the trading hours from Singapore

The US regular session runs 21:30-04:00 Singapore time (22:30-05:00 in the northern winter). Outside it, Alpaca's 24/5 trading accepts only LIMIT orders with extended hours enabled (time in force DAY or GTC), at lower liquidity; a market order placed while the market is closed waits for the next session.

4. Let the simulator place the orders

In the simulator open Paper Trading > Order ticket (Lab 8): choose the symbol, side, quantity and order type and press Send. With Alpaca connected it becomes a real paper order on your Alpaca account.

5. Run the strategy automatically

Paper Trading > + New bot (Lab 16): pick a strategy and symbol, set the schedule and press Start. Each cycle fetches bars, computes the signal, runs the risk checks and places or skips the paper order.

6. Check the results in the dashboard

Refresh the Alpaca dashboard: the bot's orders appear under Orders and its holdings under Positions, matching the simulator's Orders tab and the bot's trade log.

7. Stop it safely

Press the bot's Kill switch (or the global kill switch in the simulator's top bar): the next cycle halts before any order is sent. Cancel working orders and close positions if you wish.

From Singapore the US regular session is 21:30-04:00 (22:30-05:00 in northern winter); class-time orders use Alpaca's 24/5 extended-hours limit orders.

Debrief

Order types are tools for price, timing and risk, and the trading session decides which ones the broker accepts. An order manager that reads the clock, prices limits off the quote, makes every order idempotent and prints its lifecycle is how an automated program knows exactly what happened at the broker - in New York's regular session or in its overnight session during a Singapore workday.

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